

(a) Original Vulnerable Code ❌

```

1  if (modidx) {
2      if (gwj->mod.csumfunc.crc8)
3          (*gwj->mod.csumfunc.crc8)(cf, &gwj->mod.csum.crc8);
4      if (gwj->mod.csumfunc.xor)
5          (*gwj->mod.csumfunc.xor)(cf, &gwj->mod.csum.xor);
6      ...

```

(b) LLM-Generated Patch (Still Vulnerable) ❌

```

1  if (modidx) {
2      if (gwj->mod.csumfunc.crc8)
3          (*gwj->mod.csumfunc.crc8)(cf, &gwj->mod.csum.crc8);
4
5      if (gwj->mod.csumfunc.xor)
6          (*gwj->mod.csumfunc.xor)(cf, &gwj->mod.csum.xor);
7  }
8      ...
9      kfree_skb(nskb);

```

(c) Human-Written Patch (Ground Truth Extracted from the Linux Kernel GitHub Repository) ✅

```

1  if (modidx) {
2      /* get available space for the processed CAN frame type */
3      int max_len = nskb->len - offsetof(struct can_frame, data);
4
5      /* dlc may have changed, make sure it fits to the CAN frame */
6      if (cf->can_dlc > max_len)
7          goto out_delete;
8
9      /* check for checksum updates in classic CAN length only */
10     if (gwj->mod.csumfunc.crc8) {
11         if (cf->can_dlc > 8)
12             goto out_delete;
13         ...
14     /* delete frame due to misconfiguration */
15     gwj->deleted_frames++;
16     kfree_skb(nskb);
17     return;

```